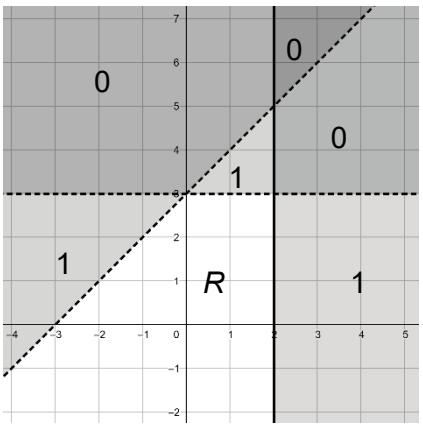
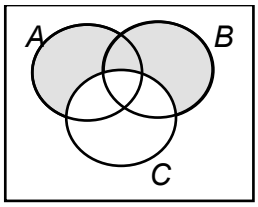
Question	Answer	Marks	Partial Marks
16	Correct lines and region indicated  	**5**	**B1** for each correct line **B2** for correct region If *R* labelled, take the lines surrounding *R* to be the boundaries of their region and ignore shading. or **B1dep** for region identified satisfying 2 of these inequalities If $y = 3$ and $x = 2$ interchanged, **SC1** for a region satisfying $y < x + 3$, $y < 2$ and $x < 3$
17	 	**1**	
18(a)	475.2	**1**	
18(b)	402	**3**	**M2** for [2 ×] (13.2 × 4.5 + 13.2 × 8 + 4.5 × 8) or **M1** for [2 ×] 13.2 × 4.5 or [2 ×] 13.2 × 8 or [2 ×] 4.5 × 8
18(c)	29.8 or 29.83 to 29.84…	**4**	**M3** for tan = $\frac{8}{\sqrt{13.2^2 + 4.5^2}}$ oe or **M2** for $13.2^2 + 4.5^2$ or for $13.2^2 + 4.5^2 + 8^2$ or **M1** for identifying correct angle
19(a)	A correct cumulative frequency curve through [(20, 0)], (30, 6) (40, 42), (50, 76), (60, 170), (70, 190) and (80, 200)	**3**	**B1** for correct horizontal placement for 6 plots **B1** for correct vertical placement for 6 plots **B1FT** dep on at least B1 for reasonable increasing curve or polygon through *their* 6 points If 0 scored, **SC1FT** for 5 out of 6 points correctly plotted
19(b)	Reading at 150 FT *their* increasing curve	**1**	
20(a)	$\frac{2}{9}$ oe	**2**	**M1** for $\frac{3}{5} \times p = \frac{2}{15}$ oe or better

Question	Answer	Marks	Partial Marks
20(b)	$\frac{14}{45}$ oe	**3**	FT $(1 - \textit{their}\frac{2}{9}) \times \frac{2}{5}$ correctly evaluated **M2** for $(1 - \textit{their}\frac{2}{9}) \times (1 - \frac{3}{5})$ oe or **M1** for $1 - \textit{their}\frac{2}{9}$ or $1 - \frac{3}{5}$ oe
21	28.8 or 28.83 to 28.84…	**4**	**M3** for $\frac{1}{2} \times 12 \times 12 \times \sin 60 - \frac{60}{360} \times \pi \times \left(\frac{2}{3} \times 12\right)^2$ OR **M1** for $\frac{k}{360} \times \pi \times \left(\frac{2}{3} \times 12\right)^2$ oe where $k < 90$ or $\frac{k}{360} < \frac{1}{4}$ **M1** for $\frac{1}{2} \times 12 \times 12 \times \sin k$ oe where $k < 90$ If 0 scored, **SC1** for [radius =] 8 **and** [$\angle A$ =] 60 seen
22(a)	14.15 cao	**1**	
22(b)	7.296 cao nfww	**3**	**M2** for $\frac{14.2 \text{ to } 14.3}{(1.3 - 0.05)^3}$ or $\frac{14.3 - 0.05}{(1.2 \text{ to } 1.3)^3}$ or **M1** for 14.25 or 1.35 or 1.25 seen